

MENGDI WANG

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EDUCATION

- TU Munich, Germany, PhD** 2022.08 - present
- **Focus:** Collaborative ML, eye tracking, XR, LLM in XR
 - **Thesis:** Preserving Eye Privacy in XR: Local ML Pipelines From Iris Obfuscation to Collaborative ML
- ETH Zurich, Switzerland, Master in Computer Science** 2019.09 - 2022.07
- **Thesis:** Exploration of Deep Features for Neural Style Transfer 📄🌐
- TU Munich, Germany, Bachelor in Informatics: Games Engineering** 2016.09 - 2019.07
- **Thesis:** Porting MLEM Algorithm for Heterogeneous Systems

EXPERIENCE

- TU Munich, Chair HCTL, Research Assistant** 2022.08 - 2026.03
- Demonstrated novel federated and split learning algorithms that essentially improve model performance and convergence in heterogeneous data distribution without incurring any extra client burden or data transfer
 - Developed novel iris recognition method leveraging style features, demonstrating superior performance and robustness compared to existing approaches, and developed the use of style transfer for iris privacy preservation
 - Built real-time LLM-driven interaction system for Pepper robot (STT + TTS + joystick control), solved compatibility issues, improved user experience, and deployed in live interaction sessions with children
 - Designed plug-and-play Unity module enabling LLM-based interaction with STT and TTS for XR apps
- ETH Zurich, Computer Graphics Lab, Research Student** 2022.02 - 2022.07
- Evaluated various factors influencing the 2D-to-3D stylization process utilizing differentiable renderer
 - Presented guidelines for style feature decomposition and controlling stroke granularity, and introduced a novel way for per-vertex color initialization to align reshaping and texturing outcomes

SELECTED PUBLICATIONS

- **CycleSL: Server-Client Cyclical Update Driven Scalable Split Learning.** *M. Wang*, E. Bozkir, E. Kasneci. Proceedings of the Winter Conference on Applications of Computer Vision (WACV), 2026 📄🌐
- **Eye-Trackd Virtual Reality: A Comprehensive Survey on Methods and Privacy Challenges.** E. Bozkir, S. Özdel, *M. Wang*, B. David-John, H. Gao, K. Butler,...E. Kasneci. Proceedings of the IEEE (PIEEE), 2026 📄
- **Iris Style Transfer: Enhancing Iris Recognition with Style Features and Privacy Preservation through Neural Style Transfer.** *M. Wang*, E. Bozkir, E. Kasneci. Proc. of the ACM on Computer Graphics and Interactive Techniques (PACMCGIT), the ACM Symposium on Eye Tracking Res. & Appl. (ETRA), 2025 📄🌐
- **Trade-Offs in Privacy-Preserving Eye Tracking through Iris Obfuscation: A Benchmarking Study.** *M. Wang*, E. Bozkir, E. Kasneci. The 25th International Conference on Digital Signal Processing (DSP), 2025 📄🌐
- **Embedding Large Language Models into Extended Reality: Opportunities and Challenges for Inclusion, Engagement, and Privacy.** E. Bozkir, S. Özdel, K. H. C. Lau, *M. Wang*, H. Gao, E. Kasneci. Proceedings of the 6th ACM Conference on Conversational User Interfaces (CUI), 2024 📄
- **TurboSVM-FL: Boosting Federated Learning through SVM Aggregation for Lazy Clients.** *M. Wang*, A. Bodonheli, E. Bozkir, E. Kasneci. The 38th AAAI Conference on Artificial Intelligence (AAAI), 2024 📄🌐

NON-RESEARCH PROJECTS

- **CloudVis:** Interactive and dynamic visualization of large-scale atmospheric data in Unity 🌐
- **SplashyWaves:** Real-time stable fluid dynamics simulation in moving container using PBD solver 🌐
- **Kings-of-Dominia:** Multi-platform touchscreen video game replicating physics-based domino toppling 🌐

OTHERS

- **Computer Skills:** Python (PyTorch), LM Studio, ComfyUI, Agent CLI, C# (Unity), CUDA, Linux, Git
- **Languages:** Chinese (native), English (C1), German (C1), Japanese (novice)
- **Teaching:** Human-AI Inter., Serious Games in XR, Recent Advances in Privacy, Intro to Prog. and Data Proc.